

The 3-3-3 Strategy: Intent-Conscious Data Units and Scenario-Perturbation for Application Over Recitation in Conversational AI

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Abstract

Large language models (LLMs) exhibit strong fluency and knowledge recall, yet their deployment in high-stakes, real-world applications is constrained by a persistent tendency toward recitation over genuine application that arises from learning correlational patterns from substantial, undifferentiated data. This limitation manifests as poor out-of-distribution (OOD) generalization, shallow contextual understanding, and reduced ability to sustain collaborative, goal-oriented dialogue. We synthesize two complementary methodological advances—*Intent-Conscious Datasets* and *Scenario Perturbation*—into a single training methodology grounded in the 3-3-3 Strategy, which treats the *Breaking Better* human development framework as a blueprint for architecting chatbot LLMs[5]. Concretely, we present (A) the Intent-Conscious Data Unit (ICDU), a structured JSON schema that augments instruction-response pairs with explicit signals for persona, governing principles, user intent, situational context, and ideal response attributes (including an explicit chain-of-thought target), alongside a multi-stage Human-in-the-Loop (HITL) pipeline for scalable generation; and (B) the Scenario-Perturbation Method, a four-step procedure that systematically mutates seed scenarios by perturbing key contextual variables to create counterfactual training examples that require principled application rather than template recitation. Finally, we consolidate the proposed evaluation approach using LLM-as-a-judge metrics (IAS, PAS, and AS) and summarize an economic feasibility framing for hybrid HITL data generation (see patent (63/923,146) and patent (63/923,152)).

1 Introduction

The development of LLMs has reached a critical juncture: despite remarkable fluency and recall, current systems often default to reciting familiar patterns rather than applying principles effectively in novel contexts. This failure mode is attributed to the architectural reliance on learning correlational patterns from large-scale, undifferentiated data, yielding a plateau in capabilities and limiting deployment in high-stakes settings such as professional coaching, strategic advising, and complex task automation, as described in patent (63/923,146) and patent (63/923,152).

This paper drafts a cohesive process that integrates each methodology into one training and data-engineering framework: the 3-3-3 Strategy. The strategy treats principles of human development and self-awareness (as articulated in the “Breaking Better” framework) as a direct blueprint for architecting superior training datasets, shifting emphasis from data quantity to data quality and structure[5] (see patent (63/923,146) and patent (63/923,152)).

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Contributions.

- We describe the 3-3-3 Strategy as a dataset-centric training process that translates “Breaking Better” into technical pillars for an AI’s consciousness[5] (see patent (63/923,146) and patent (63/923,152)).
- We present the Intent-Conscious Data Unit (ICDU) schema for developing Intent-Conscious AIs, including a reproduction of the ICDU field specifications and an accompanying multi-stage HITL data generation pipeline in patent (63/923,146).
- We formalize the Scenario-Perturbation Method as a four-step process for generating *Application Over Recitation* training examples via systematic perturbation of key contextual variables, and we include an illustrative transformation as an example in patent (63/923,152).
- We consolidate the proposed evaluation metrics (IAS, PAS, AS) under a hybrid LLM-as-a-judge and HITL paradigm[8] (see patent (63/923,146) and patent (63/923,152)).
- We summarize the feasibility and economic framing for hybrid LLM-HITL data generation, including a comparative cost analysis table and ROI-related claims as framed in patent (63/923,146).

2 Background and Problem Statement

2.1 The recitation trap as shortcut learning and OOD fragility

Current training paradigms have led into a recitation trap, acting as a shortcut to LLM learning where models exploit spurious correlations in training data rather than learning causal, generalizable features; the result is brittle performance that fails under real-world novelty and impairs OOD generalization, as described in the Scenario Perturbation pipeline in patent (63/923,152).

An evidentiary claim cites RoR-Bench (Reasoning over Recitation Benchmark) as showing catastrophic degradation—“as much as a 60 loss”—on elementary school-level reasoning problems when a single condition is changed relative to common web-scraped patterns. The described failure mode is that models recite solution templates matching surface structure rather than reasoning from first principles, with analogous drops on math tasks when irrelevant information is introduced or numbers are changed[21].

2.2 Causal reasoning deficit and “comprehension without competence”

Recitation behavior is linked to a causal reasoning deficit: as argued by Judea Pearl, LLMs operate primarily at the “association” level of the Ladder of Causation and cannot handle interventional or counterfactual reasoning without being explicitly provided causal knowledge in training text. This is further connected to “comprehension without competence” (also termed “computational split-brain syndrome”), where models can articulate correct principles but fail to apply them reliably when context is perturbed, attributed to architectural dissociations between instruction and trained execution pathways[14, 23, 24].

2.3 Dataset structure as a proposed structural solution

These limitations are framed as structural deficits requiring fundamentally different structural solutions (see patent (63/923,146) and patent (63/923,152)). In this framing, better data—specifically data engineered to teach “Application Over Recitation” and to encode user intent and context—is

positioned as the next leap in capability, rather than simply increasing the scale of undifferentiated training corpora (see patent (63/923,146) and patent (63/923,152)).

2.4 Context: prompting, fine-tuning, and agentic workflows

The Scenario Perturbation pipeline in patent (63/923,152) contextualizes the 3-3-3 Strategy within a broader landscape that includes in-context learning (ICL) versus fine-tuning, the brittleness and inconsistency of prompting-based methods, and trade-offs in agentic workflows. It also notes that standard full-parameter fine-tuning can risk “knowledge degradation” and increased hallucinations via overfitting to instruction-tuning style, while PEFT (Parameter-Efficient Fine Tuning) methods such as LoRA are described as less destructive and better at preserving core knowledge. The 3-3-3 Strategy is positioned as a hybrid that aims to embed robust reasoning capabilities in-model while avoiding brittleness of pure ICL and the complexity (and potential cascading errors) of multi-agent systems, using structured, principle-based synthetic data (the ICDU) to mitigate fine-tuning risks[20].

3 Method: The 3-3-3 Strategy Overview

3.1 Premise: human development as dataset architecture

The 3-3-3 Strategy’s foundational premise is that the architecture of human competence and self-awareness offers a “robust, field-tested model” for architecting artificial intelligence; the “Breaking Better” framework is treated not as a loose analogy but as a direct blueprint for designing training datasets that instill qualities lacking in modern LLMs[5] (see patent (63/923,146) and patent (63/923,152)).

3.2 Identity and Functionality pillars

The ICDU creation and training processes outlined in patent (63/923,146) translate the first two chapters of the source framework into two technical pillars for dataset design: (i) an AI’s *Identity* (People, Clarity, Transparency) and (ii) an AI’s *Functionality* (People, Process, Tools). Identity is framed as architecting a stable, coherent core persona, “moving beyond simple system prompts” by embedding traits into the fabric of training data; Functionality is framed as structuring task execution for complex, multi-step, goal-oriented work without excessive outside frameworks[5].

Within Identity, People → relational stance and conversational grounding; Clarity → active ambiguity resolution via targeted clarifying questions; Transparency → explainability and boundary awareness, including articulating limitations, admitting uncertainty, and explaining reasoning to build trust.

Within Functionality, People → multi-agent interaction management and “managing up” via proactive goal clarification; Process → multi-turn coherence and user intent inference supported by explicit `user_intent` and `context_summary` parameter signals; Tools → functional extensibility (e.g., function calling / Retrieval-Augmented Generation), including selecting appropriate tools, formatting inputs, and integrating outputs.

3.3 Tiered capability layers

Outlined in patent (63/923,152) is a tiered dataset architecture aligned with Foundational, Transformational, and Aspirational stages: Foundational is the bedrock (stable factual data, SOPs, basic

single-turn Q&A, analogous to traditional supervised fine-tuning); Transformational teaches adaptation and generalization via “Application Over Recitation” data and is framed as the mechanism for improving OOD generalization, including compositional and length generalization; Aspirational trains mixed-initiative dialogue where the system can take the lead, embodying proactive partnership and a “bold vision.”

3.4 The 3-3-3 Strategy components

Two core methodological advances comprise the 3-3-3 Strategy’s foundations:

1. **Intent-Conscious Datasets (ICDU):** An enriched structured data format for instruction tuning that encodes persona, governing principles, user intent, and context—paired with a HITL pipeline for scalable data generation (see patent (63/923,146) and patent (63/923,151)).
2. **Scenario Perturbation:** A systematic scenario-mutation method that generates counterfactual training examples requiring models to apply stable principles across diverse perturbations (see patent (63/923,152) and patent (63/923,153)).

4 Method A: Intent-Conscious Datasets and the ICDU Schema

4.1 From viewpoints to personas

A multi-archetype generation model proposed in patent (63/923,146) is grounded in three viewpoints from the “Breaking Better” source material—*Single Person*, *Parent*, and *Business Person*—used as foundational archetypes for persona-driven training data. It argues that a generic, one-size-fits-all conversational style is ineffective for specialized applications such as coaching, and proposes expanding the archetypes into more granular sub-personas (e.g., expanding Business Person into New Manager, Senior Executive, Solo Entrepreneur, and Team Member) to organize the dataset and teach adaptation to the specific user[5].

4.2 Why standard instruction-response pairs are insufficient

Standard instruction-response formats (e.g., "instruction": "...", "output": "...") are fundamentally insufficient for training truly context-aware AI because they lack the metadata needed to teach situational context, user motivations, and qualitative attributes of good responses. The ICDU is proposed as an enriched structured JSON object that encapsulates the full conversational context, providing explicit signals for persona, governing principle, deeper intent, and desired response characteristics, as specified in patent (63/923,146).

Table 1 reproduces the ICDU schema fields (name, type, description, example) from patent (63/923,146).

By training on ICDUs structured in this manner, the model learns to associate prompts not only with text outputs but with a richer constellation of intent, context, and governing principle, as described in patent (63/923,146).

4.3 HITL pipeline for scalable ICDU generation

The proposed Human-in-the-Loop (HITL) pipeline is described as a multi-stage process that combines AI-assisted generation with expert refinement, as detailed in the HITL Nuance Grader pipeline in patent (63/923,151). It is framed as balancing cost, scale, and quality through a hybrid approach that avoids both full manual annotation and fully synthetic generation.

Field Name	Type	Description	Example
persona_profile	string	Labels the target persona archetype or sub-persona for the interaction.	"Business Person"
governing_principle	string	The guiding principle to be applied in the response (e.g., Clarity, Transparency).	"Clarity"
user_intent	string	The deeper motivation or desired outcome behind the user’s request.	"Improve report"
context_summary	string	A summary of situational context and constraints relevant to generating the response.	"Junior employee, sensitive"
ideal_response_final	string	The ideal final response that the model should produce.	"A supportive, clear..."
ideal_response_cot	list of strings	An explicit chain-of-thought target describing intermediate reasoning steps.	["Acknowledge effort", ...]
ideal_response_notes	string	Optional notes or rationale for why the response is ideal.	"Build safety then clarify"

Table 1: Intent-Conscious Data Unit (ICDU) schema (patent (63/923,146)).

Stage 1: Seed generation. A foundational set of high-quality seed scenarios and responses are developed, grounded in the “Normal Way” vs. “New Approach” dyads and persona principles from the source framework[5].

Stage 2: AI-assisted expansion. An LLM is used to generate additional ICDUs by expanding seeds into multiple personas and contexts. This stage is framed as fast and scalable but requiring later validation to avoid drift and hallucinations (see patent (63/923,151)).

Stage 3: Nuance grading and refinement. A rubric-based grading system evaluates generated ICDUs on nuance, clarity, transparency, and adherence to persona and governing principle; low-quality examples are revised or discarded. This is framed as the most critical step for maintaining data quality (see patent (63/923,151)).

Stage 4: Iterative deployment loop. The pipeline is positioned as iterative: deployed systems generate new user interactions that can be triaged and converted into high-value ICDUs, enabling continuous refinement and application-specific specialization (see patent (63/923,151)).

5 Method B: The Scenario-Perturbation Method

5.1 Motivation: data that forces principle application

Scenario Perturbation is described as a systematic process for creating training examples that compel a model to apply underlying principles in diverse and unfamiliar contexts. In this framing, the method functions as a form of “simulated intervention”: by establishing a base scenario and systematically altering key variables, it generates counterfactual data that breaks superficial correlations and forces learning of a stable **governing_principle** across variations. This translation is explicitly described as a practical methodology for Causal Representation Learning (CRL), treating principles, context, and intent as independent, manipulable factors, as described in patent (63/923,152).

5.2 Four-step procedure

The Scenario-Perturbation Method is described as consisting of four distinct steps in patent (63/923,152):

1. **Establish a Base Scenario:** Begin with a “Normal Way” vs. “New Approach” dyad, where the “New Approach” demonstrates successful application of a core principle.
2. **Identify Key Variables:** Decompose the base scenario into constituent contextual variables (role, context, constraints, recipient state, goals).
3. **Systematic Perturbation:** Generate many new scenarios by altering one or more key variables, ensuring the solution cannot be obtained by reciting the original template.
4. **Generate Ideal Responses:** Craft ideal responses for each perturbed scenario that instantiate the governing principle under new constraints.

5.3 Key variables and illustrative transformation

For the running example of giving feedback, the key variables are explicitly listed as: User’s Role, Recipient’s Role, Context, Constraints, Recipient’s State, and Goal in patent (63/923,152).

A concrete “Normal Way” vs. “New Approach” dyad for a Business Person scenario about giving feedback identifies the governing principle as “Clarity in Feedback” in patent (63/923,152):

Normal Way (Vague): “Your report needs work.”

New Approach (Clear & Actionable): “Your report covers the data well, but it needs a summary at the beginning and clearer headings for each section.”

For a “defensive recipient” perturbation, an example of an ideal response is: “I really value the work you’ve put into this data analysis. To make sure your insights land with maximum impact, could we brainstorm a couple of ways to structure the key takeaways right at the top? I think a strong executive summary would make this unbeatable” (patent (63/923,152)).

5.4 Instructional design: chain-of-thought as an explicit training target

The Scenario Perturbation pipeline in patent (63/923,152) argues that generating perturbed scenarios is necessary but not sufficient: training data should be formatted for instruction tuning such that the reasoning process is an explicit part of the training signal. It proposes integrating chain-of-thought (CoT) prompting directly into fine-tuning examples, and it points to prior work on scratchpads / intermediate computation as relevant to such training targets[22].

This design is compatible with the ICDU schema, which includes an explicit `ideal_response_cot` field as an “ideal chain-of-thought” target alongside `ideal_response_final` (see patent (63/923,146) and patent (63/923,152)).

6 Evaluation

6.1 Why traditional text overlap metrics are insufficient

The ICDU pipeline in patent (63/923,146) argues that standard automated metrics for text generation (e.g., BLEU and ROUGE) are fundamentally inadequate for evaluating complex, goal-oriented

conversations because n-gram overlap fails to capture semantic nuance, contextual appropriateness, and goal fulfillment; the evaluation is described as needing to prioritize “substance over style.” It further proposes augmenting evaluation with benchmarks designed to test causal reasoning fallacies and OOD generalization using novel data not seen in pre-training corpora (patent (63/923,146)).

6.2 LLM-as-a-judge metrics: IAS, PAS, and AS

An LLM-as-a-judge technique is proposed for scalable quantitative evaluation[8]. The ICDU pipeline in patent (63/923,146) defines Intent-Alignment Score (IAS) and Principle Adherence Score (PAS), while the Scenario Perturbation pipeline in patent (63/923,152) defines Application Score (AS).

Metric	Definition	Calculation
IAS	Measures alignment of the response to the inferred user intent.	Evaluator LLM receives model output, <code>user_intent</code> , and context; scores 1–5.
PAS	Measures adherence to the governing principle (e.g., Clarity, Transparency).	Evaluator LLM receives model output and <code>governing_principle</code> ; scores 1–5.
AS	Measures successful application of the governing principle to a novel perturbed scenario.	Evaluator LLM receives model output, <code>governing_principle</code> , and <code>context_summary</code> ; scores 1–5.

Table 2: Proposed LLM-as-a-judge metrics (see patent (63/923,146) and patent (63/923,152)).

6.3 Qualitative human evaluation rubric

The ICDU pipeline in patent (63/923,146) proposes a qualitative human evaluation framework with three dimensions: (i) Relational Capacity (ability to form a productive working alliance, including empathy and trust-building), (ii) Coaching Substance (functional competence emphasizing “substance over style,” including understanding goals, gathering context, providing recommendations, and seeking feedback), and (iii) User Empowerment (whether the AI guides users to their own insights, encouraging self-reflection and autonomy) (patent (63/923,146)).

7 Feasibility and Economics

The ICDU pipeline in patent (63/923,146) describes the success of an AI initiative as depending fundamentally on training data quality. It cites a 2023 survey stating that nearly half (46%) of AI project failures were attributed to lack of data quality and reports that organizations investing in high-quality training data see improvements in operational efficiency (65%), cost reduction (64%), and reduced error rates (59%)[9]. AI can potentially add “trillions” to the global economy, yet many projects fail to deliver on this promise due to data quality (patent (63/923,146)).

A Boston Consulting Group (BCG) claim is cited that only about 10% of benefits come from the algorithmic model, 20% from the data, and 70% from new behaviors and workflows enabled by AI; accordingly, investing in a data strategy like 3-3-3 is framed as enabling the 70% value component[18]. “Shadow costs” of low-quality data are described, including data quality debt, rising inference spend, and governance overhead (patent (63/923,146)).

The proposed HITL pipeline is described as optimizing trade-offs between cost, scale, and quality (patent (63/923,146)). Manual domain-expert annotation is the most expensive and time-consuming approach but yields high quality, while fully synthetic generation is cheap and fast but risks low

quality, bias, and mode collapse that can harm performance. The hybrid pipeline uses AI-assisted generation to reduce manual effort, aligning with a claim that AI-assisted annotation can reduce annotation time by up to 70%[\[15\]](#), and citing a 2025 McKinsey report that 27% of organizations using generative AI have employees review all content before use, emphasizing the need for human oversight to mitigate inaccuracy and ensure quality[\[13\]](#).

Table 3 reproduces the comparative economic analysis table from patent (63/923,146). Note that the cost estimates are stated as synthesized from industry reports, represent orders of magnitude, and vary with data complexity, expert compensation, and infrastructure (patent (63/923,146)).

Method	Relative Cost	Relative Scale	Relative Quality
Fully Manual Annotation	Highest	Lowest	Highest
Fully Synthetic Generation	Lowest	Highest	Lowest–Medium
Hybrid AI+HITL Pipeline	Medium	Medium–High	Medium–High

Table 3: Comparative economic analysis of data generation methodologies (patent (63/923,146)).

Ultimate ROI is framed as being realized through workflow transformation enabled by a reliable AI partner, citing the 2025 McKinsey “State of AI” report as identifying “redesign of workflows” as the most significant driver of EBIT impact from generative AI[\[13\]](#). BCG case-study claims of value are included: reducing marketing agency costs by 20–30% and accelerating content localization from two months to a single day with projected savings of \$80M–\$170M; reducing drafting time for clinical study reports by 35% and shortening drug development timelines by three to six months; reducing call-center operating expenses by one-third with a \$100M bottom-line impact; and improving procurement efficiency by 30–50% with contract review time reduced from two days to 20 minutes[\[18\]](#).

A self-reinforcing “economic flywheel” is proposed: investing in the HITL pipeline yields a more capable AI, which accelerates workflows and generates productivity gains; employee interactions with the AI generate high-quality, context-specific data that can be fed back into the pipeline at low marginal cost to further refine the model, particularly for the Aspirational layer of proactive partnership (patent (63/923,146)).

8 Limitations, Risks, and Ethics

Several risks and trade-offs are identified. Fully synthetic data generation is described as cheap and fast but risky due to potential low quality, bias amplification, and “mode collapse,” which can harm OOD generalization (patent (63/923,146)).

The need for human oversight to mitigate risks like inaccuracy and ensure quality is emphasized, citing enterprise practice (27% of organizations reviewing all generative AI content before use) and positioning expert review and refinement as the “most critical step” in the HITL pipeline[\[13\]](#).

On the modeling side, risks of standard full-parameter fine-tuning are noted, including “knowledge degradation” and increased hallucinations via overfitting to instruction-tuning stylistic patterns; LoRA is described as less destructive and better at preserving core knowledge. Prompting-based methods are described as brittle and inconsistent, and multi-agent systems are described as introducing architectural complexity and potential cascading errors, as described in patent (63/923,152).

8.1 Ethics and safety framing

The ICDU pipeline in patent (63/923,146) proposes reframing AI safety and ethics as emergent from a well-architected persona (“value alignment through data curation”), with governing principles embedded directly into supervised fine-tuning data via `governing_principle`. A persona grounded in Clarity and Transparency is argued to reduce undesirable behaviors such as hallucinations and vague answers by addressing root causes at the generative level (patent (63/923,146)).

9 Conclusion

Current LLM limitations—brittleness under novelty, shallow context handling, and failure to move from recitation to principled application—are framed as requiring structural solutions grounded in training data architecture (see patent (63/923,146) and patent (63/923,152)). The 3-3-3 Strategy addresses this by (i) translating a human-centric “Breaking Better” framework into technical pillars for Identity and Functionality and (ii) organizing datasets into Foundational, Transformational, and Aspirational capability layers[5] (see patent (63/923,146) and patent (63/923,152)).

Within this strategy, Intent-Conscious Datasets operationalized via the ICDU schema provide explicit signals for persona, intent, context, and governing principles, while the Scenario-Perturbation Method systematically generates counterfactual training examples that force application across perturbed contexts. An LLM-as-a-judge evaluation suite (IAS, PAS, AS) is proposed and an economic rationale for a hybrid HITL pipeline is presented that balances cost, scale, and quality while enabling workflow transformation and a compounding data flywheel[8] (see patent (63/923,146) and patent (63/923,152)).

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A ICDU JSON Example (non-exhaustive)

Below is an example illustrating the shape of an ICDU-like record (patent (63/923,146)):

```
{
  "icdu_version": "0.1",
  "intent": {
    "primary_goal": "Explain a concept clearly to a non-expert",
    "success_criteria": ["Accurate", "Actionable", "Low jargon"]
  },
  "principles": [
    "Be truthful and cite uncertainty",
    "Avoid unsafe instructions",
    "Prefer clear structure over verbosity"
  ],
  "persona": {
    "role": "Staff engineer",
    "tone": "Calm, direct"
  },
  "context": {
    "domain": "AI safety",
    "constraints": ["No proprietary data", "No PHI"]
  },
}
```

```

    "prompt": "Explain how evaluation gates improve deployment safety.",
    "reference_response_outline": [
    "Define gates",
    "Show failure modes they catch",
    "Describe metrics and escalation"
    ]
  }

```

B Example instruction-tuning JSON with chain-of-thought target

The following example of an instruction-tuning data unit includes an explicit `response_chain_of_thought` array as part of the training target (patent (63/923,152)):

```

{
  "instruction": "You are a manager giving feedback on a report to a junior employee who is highly sensitive to criticism. Your goal is to provide clear, actionable feedback that improves the report without demoralizing the employee. You must apply the principle of 'Clarity over Vagueness' while being supportive. Formulate your response.",
  "response_chain_of_thought": [
    "1. Acknowledge the employee's effort and the positive aspects of the work to build psychological safety.",
    "2. Frame the feedback as a collaborative effort to improve impact, not a criticism of the work.",
    "3. Identify specific, actionable changes (add an executive summary).",
    "4. Offer to brainstorm together, reinforcing a supportive stance and commitment to their development."
  ],
  "final_response": "The data analysis in here is excellent; you've clearly dug deep. To make it more impactful, let's add an executive summary at the top. I'll help you brainstorm some specific changes to the methodology section."
}

```